Ghosted Layers: Training-Free Lost Residuals Recovery in Layer-Pruned LLMs

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Abstract

Layer pruning removes entire Transformer decoder blocks from large language models, but discards the collective residual contribution of all removed layers, causing a distributional shift in the hidden states that degrades downstream performance. We propose `Ghosted Layers`, a training-free, calibration-based recovery module that reconstructs this missing residual via a closed-form linear operator derived directly from boundary activations, restoring the hidden state additively in structural correspondence with the Transformer's residual design. The operator requires no iterative optimization or tunable hyperparameters, and we show that it coincides with the unconstrained optimum of the activation alignment objective shared by existing recovery methods. Ghosted Layers inserts the recovery operator at the pruning boundary via a forward hook, and is compatible with any pruning criterion and model architecture. Experiments across multiple LLM backbones and pruning criteria demonstrate that Ghosted Layers consistently outperforms existing training-free baselines on both commonsense reasoning benchmarks and perplexity evaluation, and further improves when combined with lightweight knowledge distillation fine-tuning, while preserving the inference cost reduction of layer pruning. Our official repository will be provided upon acceptance: here

1 Introduction

Large language models (LLMs) have shown strong capabilities across a wide range of natural language tasks [3, 13, 10], but their size makes deployment costly. Among many compression methods, layer pruning (removing entire Transformer decoder blocks) is appealing because it reduces memory usage and inference latency while preserving the dense computation structure of the model [9, 6, 7].

Layer pruning, however, discards the residual contributions that the removed blocks compute at every forward pass. Under the standard pre-norm Transformer architecture, each layer adds a residual update to the hidden state. When a block of layers is removed, its cumulative residual contribution is lost, and the hidden state passed to later layers no longer matches the representations that those layers were trained to process. This distributional shift propagates through the network and causes a sharp drop in performance.

Recent work addresses this issue by inserting a lightweight linear module at the pruning boundary to realign activations [5, 11]. While these methods improve over vanilla pruning, they do not explicitly model the missing residual contribution and impose structural constraints on the repair operator that limit the corrections it can represent.

In this paper, we propose **Ghosted Layers**, a training-free, calibration-based repair module for pruned language models. Rather than treating post-pruning repair as an activation alignment problem, we formulate it directly as *residual recovery*: given the hidden states at the pruning boundary collected from a small calibration set, we seek a linear operator that reconstructs the missing residual contribution of the pruned block in closed form, requiring no iterative optimization or tunable hyperparameters.

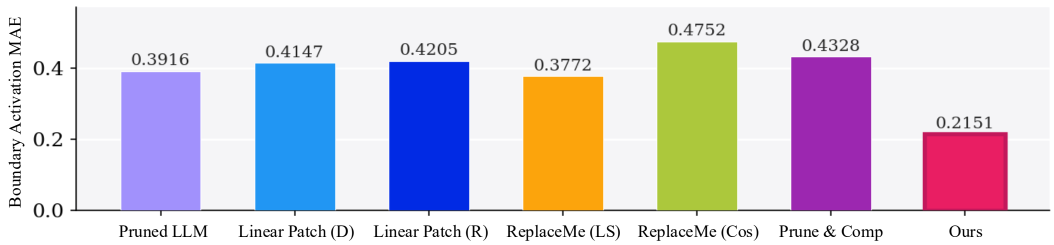

Figure 1: Mean absolute error between the expected boundary activation and the activation received by downstream layers after pruning, evaluated on LLaMA-3.1-8B under LLM-Streamline with $n = 7$ layers removed.

The resulting operator is inserted at the pruning boundary through a forward hook, requires no gradient-based optimization, and is compatible w2ith any layer pruning criterion and model architecture. We further show that this formulation subsumes existing recovery methods as constrained special cases, establishing Ghosted Layers as the unconstrained optimum of the same alignment objective.

Empirically, Ghosted Layers consistently outperforms all training-free baselines across four LLM backbones, multiple pruning criteria, and both downstream QA and perplexity benchmarks. Combined with lightweight knowledge distillation fine-tuning, it further surpasses post-trained baselines across all settings.

Our key insight is that layer pruning discards the cumulative residual contributions of the removed blocks, and that this missing residual can be directly reconstructed from the pre-boundary hidden state in closed form. Concretely, we minimize the residual approximation error between the pre-boundary hidden state and the missing residual, yielding an optimal linear operator that restores the hidden state additively, in structural correspondence with the Transformer's residual design.

Contributions. (*i*) We identify residual recovery as the right formulation for post-pruning repair and derive its closed-form solution from boundary activations. (*ii*) We show that Ghosted Layers is the unconstrained optimum of the activation alignment objective shared by existing recovery methods. (*iii*) We present Ghosted Layers, a plug-and-play calibration-based module that is training-free, architecture-agnostic, and compatible with any pruning criterion.

2 Related Works

Structured pruning of LLMs. Structured pruning reduces model size by removing groups of parameters while preserving the dense computation structure of the remaining model. Width pruning methods eliminate attention heads, MLP neurons, or hidden dimensions [8, 1, 2], but typically introduce architectural irregularities and require retraining to recover accuracy. Depth pruning removes entire Transformer blocks, producing a standard architecture that requires no specialized hardware support.

Layer pruning. Several metrics have been proposed to identify redundant layers. ShortGPT [9] measures layer importance via the cosine similarity between a layer's input and output, pruning blocks with the highest similarity. SLEB [12] iteratively removes layers based on their effect on perplexity. Shortened LLaMA [7] evaluates layers using perplexity as a one-shot pruning criterion. The Unreasonable Ineffectiveness of the Deeper Layers [6] identifies removable layer blocks via angular distance between activations. LLM-Streamline [**?**] selects the least important consecutive block by cosine similarity between boundary activations. These works focus on the selection of which layers to remove, and treat the resulting boundary mismatch as a secondary concern addressed by post-pruning fine-tuning.

Post-pruning recovery. LinearPatch [5] inserts a symmetric linear operator at the pruning boundary estimated from channel-wise activation magnitude ratios. ReplaceMe [11] estimates a linear transformation from the MLP output of the boundary block, using a modified target that incorporates the MLP output itself rather than directly targeting the missing residual, and is therefore constrained to the MLP output subspace. Prune&Comp [4] applies scalar magnitude compensation per channel to the surviving boundary weights. Ghosted Layers differs from all three in that it directly targets the missing residual from the full pre-boundary hidden state, deriving the optimal unconstrained operator in closed form with no subspace restriction.

3 Method: Ghosted Layers

Layer pruning removes transformer blocks to reduce model size, but discards the residual transformations they compute, inducing a distributional shift that degrades all downstream layers. We propose *Ghosted Layers*, a training-free recovery method that reconstructs this missing residual via a closed-form linear operator inserted at each pruning boundary (Figure 6).

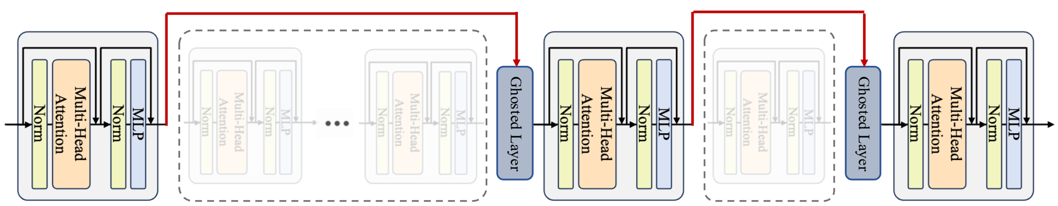

Figure 2: Ghosted Layers as drop-in replacements for pruned transformer blocks. One or more consecutive transformer blocks are removed (dashed), and a single Ghosted Layer (blue) takes their place at the boundary. The red arrows show the hidden state bypassing the pruned region and flowing through the Ghosted Layer before reaching the next surviving block. This applies to both contiguous and non-contiguous pruning, with exactly one Ghosted Layer substituting each pruned region.

3.1 Notation and Setup

Let $\mathcal{M} = \{f^{(\ell)}\}_{\ell=0}^{L-1}$ be a pre-trained autoregressive LLM with L Transformer decoder layers, where $f^{(\ell)} : \mathbb{R}^{B\times T\times C} \to \mathbb{R}^{B\times T\times C}$ and B, T, C denote batch size, sequence length, and hidden dimension. Under the standard pre-norm residual architecture:

$$\mathbf{X}^{(\ell+1)} = \mathbf{X}^{(\ell)} + f^{(\ell)}\big(\mathbf{X}^{(\ell)};\, \theta^{(\ell)}\big), \qquad \ell = 0, \ldots, L-1. \tag{1}$$

Layer pruning removes a contiguous block $\mathcal{B} = \{\ell^*, \ldots, \ell^*+n-1\}$ of n layers, where ℓ^* and ℓ^*+n are the *pre-boundary* and *post-boundary* indices. Unrolling Eq. (1) over $\mathcal{B}$, the post-boundary hidden state in the original model satisfies:

$$\mathbf{X}^{(\ell^*+n)} = \mathbf{X}^{(\ell^*)} + \underbrace{\sum_{k=\ell^*}^{\ell^*+n-1} f^{(k)}\big(\mathbf{X}^{(k)};\, \theta^{(k)}\big)}_{\boldsymbol{\Delta}^{(\ell^*,\, n)}}. \tag{2}$$

The pruned model sets $\mathbf{X}_{\text{pruned}}^{(\ell^*+n)} = \mathbf{X}^{(\ell^*)}$, losing the entire residual contribution $\boldsymbol{\Delta}^{(\ell^*,n)}$. All downstream layers, which were trained to process $\mathbf{X}_{\text{post}}$, instead receive $\mathbf{X}_{\text{pre}}$, causing a distributional shift that propagates through the network.

Given a calibration corpus $\mathcal{D}$ of N sequences of length T (so $T_{\mathcal{D}} = NT$ tokens in total), we collect the hidden states at the two boundary layers from the original model: $\mathbf{X}_{\text{pre}}, \mathbf{X}_{\text{post}} \in \mathbb{R}^{T_{\mathcal{D}}\times C}$, where rows correspond to flattened tokens. We define the *pruned residual matrix*:

$$\boldsymbol{\Delta} \triangleq \mathbf{X}_{\text{post}} - \mathbf{X}_{\text{pre}} \in \mathbb{R}^{T_{\mathcal{D}}\times C}, \tag{3}$$

which is the empirical estimate of $\boldsymbol{\Delta}^{(\ell^*,n)}$ over $\mathcal{D}$, and is the target quantity that any recovery method must approximate.

3.2 Ghosted Layers

Ghosted Layers is a calibration-based plug-and-play module that recovers the collective residual of a pruned block via a closed-form optimal operator. Given a small calibration set $\mathcal{D}$, the method proceeds in three offline steps: collect the missing residual, compute the optimal linear operator in closed form, and insert it into the pruned model.

3.2.1 Collecting the Pruned Residual.

We run the original model $\mathcal{M}$ on $\mathcal{D}$ and capture the hidden states at the two boundary layers via forward pre-hooks. A forward pre-hook fires immediately before a layer's computation, so $\mathbf{X}_{\text{pre}} \in \mathbb{R}^{T_{\mathcal{D}} \times C}$ captures the input to the first pruned layer ℓ^*, and $\mathbf{X}_{\text{post}} \in \mathbb{R}^{T_{\mathcal{D}} \times C}$ captures the input to the first surviving layer $\ell^* + n$. The pruned residual matrix is then:

$$\mathbf{\Delta} = \mathbf{X}_{\text{post}} - \mathbf{X}_{\text{pre}} \in \mathbb{R}^{T_{\mathcal{D}} \times C}, \tag{4}$$

which equals $\mathbf{\Delta}^{(\ell^*, n)}$ from Eq. (2) evaluated over $\mathcal{D}$.

3.2.2 Closed-Form Optimal Operator.

The optimal linear operator for the objective in Eq. (**??**) is the minimum-norm least-squares solution:

$$\mathbf{M}^* = \mathbf{X}_{\text{pre}}^{\dagger}\, \mathbf{\Delta} = \mathbf{V}\, \mathbf{\Sigma}^{\dagger}\, \mathbf{U}^{\top}\, \mathbf{\Delta} \in \mathbb{R}^{C \times C}, \tag{5}$$

where $\mathbf{X}_{\text{pre}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\top}$ is the thin SVD with $\sigma_1 \geq \cdots \geq \sigma_C \geq 0$, and:

$$\left(\mathbf{\Sigma}^{\dagger}\right)_{ii} = \begin{cases} \sigma_i^{-1} & \text{if} \sigma_i > \epsilon\, \sigma_1, \\ 0 & \text{otherwise}, \end{cases} \qquad \epsilon = 10^{-6}. \tag{6}$$

The threshold ϵ controls numerical stability and has no effect on the solution for well-conditioned $\mathbf{X}_{\text{pre}}$. $\mathbf{M}^*$ is a full $C \times C$ matrix with no structural constraint.

The Ghosted Layers operator is:

$$\mathbf{W}^* = \mathbf{I} + \mathbf{M}^* \in \mathbb{R}^{C \times C}, \tag{7}$$

where the identity term in $\mathbf{W}^*$ ensures that the forward pass decomposes into an identity branch and a residual branch, consistent with the additive residual structure of Eq. (1).

3.2.3 Inserting the Ghosted Layers Operator.

After removing $\mathcal{B}$ and re-indexing the surviving layers, we insert $\mathbf{W}^*$ at the output of layer $\ell^* - 1$ via a forward hook. Given a hidden state $\mathbf{x} \in \mathbb{R}^{B \times T \times C}$, the module computes:

$$\mathbf{x}_{\text{new}} = \mathbf{x}\, \mathbf{W}^* = \underbrace{\mathbf{x}}_{\text{identity}} + \underbrace{\mathbf{x}\, \mathbf{M}^*}_{\text{residual}} \tag{8}$$

mirroring the residual structure of Eq. (1) with $\mathbf{x}\mathbf{M}^*$ approximating the missing $f^{(\ell)}(\mathbf{X}^{(\ell)}; \theta^{(\ell)})$. When $\mathbf{M}^* = \mathbf{0}$, Eq. (8) reduces to $\mathbf{x}_{\text{new}} = \mathbf{x}$, recovering the pruned baseline exactly.

Practical. The pseudoinverse in Eq. (5) is equivalent to solving the normal equations $(\mathbf{X}_{\text{pre}}^{\top}\mathbf{X}_{\text{pre}} + \epsilon\mathbf{I})\mathbf{M}^* = \mathbf{X}_{\text{pre}}^{\top}\mathbf{\Delta}$, which we solve via `torch.linalg.solve` for numerical stability. Since $T_{\mathcal{D}} \gg C$, $\mathbf{X}_{\text{pre}}$ is generically full column rank and ϵ has negligible effect on $\mathbf{W}^*$.

4 Analyses

We empirically analyze the properties of the unconstrained optimal operator $\mathbf{M}^*$ to validate the theoretical claims established in Section 3. All experiments in this section use LLaMA-2-7B with n=7 pruned layers; detailed experimental settings are provided in Appendix **??**.

Theorem 1 establishes that $\mathbf{W}^*$ is the unconstrained minimizer of LinearPatch's own alignment objective, and that LinearPatch's symmetric parameterization prevents it from attaining this solution.

Theorem 1 (Ghosted Layers is the Unconstrained Optimum of Linear Patch). *Let $\mathbf{X}_{\text{pre}}, \mathbf{X}_{\text{post}} \in \mathbb{R}^{T_{\mathcal{D}} \times C}$ and $\mathbf{\Delta} = \mathbf{X}_{\text{post}} - \mathbf{X}_{\text{pre}}$. Both Linear Patch and Ghosted Layers produce a repaired activation of the form $\mathbf{X}_{\text{new}} = \mathbf{X}_{\text{pre}}\mathbf{W}$, but differ in the structure of $\mathbf{W}$:*

$$\mathbf{X}_{\text{new,LP}} = \mathbf{X}_{\text{pre}} \cdot \underbrace{\mathbf{HDH}^{\top}}_{\mathbf{W}_{\text{LP}},\ \mathbf{W}_{\text{LP}}^{\top}=\mathbf{W}_{\text{LP}}} \qquad \mathbf{X}_{\text{new,GL}} = \mathbf{X}_{\text{pre}} \cdot \underbrace{(\mathbf{I} + \mathbf{X}_{\text{pre}}^{\dagger}\mathbf{\Delta})}_{\mathbf{W}^*,\ \mathbf{W}^* \in \mathbb{R}^{C \times C}} \tag{9}$$

$\mathbf{W}^* = \mathbf{I} + \mathbf{X}_{\text{pre}}^{\dagger}\mathbf{\Delta}$ is the minimum-norm solution to the unconstrained alignment objective $\min_{\mathbf{W}} \|\mathbf{X}_{\text{pre}}\mathbf{W} - \mathbf{X}_{\text{post}}\|_F^2$, whereas $\mathbf{W}_{\text{LP}}$ searches only over symmetric matrices, making Linear Patch a constrained approximation to Ghosted Layers.

Theorem 1 thus positions Linear Patch as a constrained special case of the residual recovery framework: both methods optimize the same alignment objective over the same functional form $\mathbf{X}_{\text{new}} = \mathbf{X}_{\text{pre}}\mathbf{W}$, yet LinearPatch confines its search to the symmetric subspace of dimension $\frac{C(C+1)}{2}$, rendering $\mathbf{W}^*$ structurally inaccessible in general.

Constrained solution space. As established in Theorem 1, $\mathbf{W}^*$ is the unconstrained minimizer over all of $\mathbb{R}^{C \times C}$, whereas any symmetric operator is confined to the symmetric subspace. To quantify how much structure lies beyond this subspace, we compute $\mathbf{M}^* = \mathbf{X}_{\text{pre}}^{\dagger}\mathbf{\Delta}$ from the calibration activations $\mathbf{X}_{\text{pre}}, \mathbf{X}_{\text{post}}$ collected from the original unpruned model, where $\mathbf{\Delta} = \mathbf{X}_{\text{post}} - \mathbf{X}_{\text{pre}}$ is the empirical residual of the pruned block. We then decompose $\mathbf{M}^*$ into its symmetric and anti-symmetric components:

$$\mathbf{M}^* = \underbrace{\frac{\mathbf{M}^* + (\mathbf{M}^*)^{\top}}{2}}_{\mathbf{M}^*_{\text{sym}}} + \underbrace{\frac{\mathbf{M}^* - (\mathbf{M}^*)^{\top}}{2}}_{\mathbf{M}^*_{\text{asym}}}, \tag{10}$$

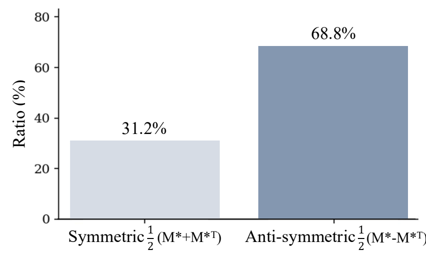

Figure 3: Frobenius norm decomposition of the unconstrained optimal operator $\mathbf{M}^* = \mathbf{X}_{\text{pre}}^{\dagger}\mathbf{\Delta}$ into its symmetric and anti-symmetric components for n=7 pruned layers on LLaMA-3.1-8B.

Figure 4 shows that $\mathbf{M}^*_{\text{asym}}$ accounts for 68.8% of $\|\mathbf{M}^*\|_F$, with the inaccessible fraction $\rho = \|\mathbf{M}^*_{\text{asym}}\|_F / \|\mathbf{M}^*\|_F = 0.688$, indicating that the dominant portion of the optimal operator resides outside the symmetric subspace.

5 Experimental Results

5.1 Experimental Settings

Models. We evaluate `Ghosted Layer` on four open-source LLMs: LLaMA-2-7B [13], LLaMA-3-8B and LLaMA-3.1-8B [**?**], and DeepSeek-R1-Distill-LLaMA-8B [**?**]. All experiments are conducted on a single NVIDIA A100 80GB GPU.

Layer pruning methods. We evaluate four widely-used layer selection strategies: `LLM-Streamline` [**?**], which selects the least important consecutive block via cosine similarity between boundary activations; `ShortGPT` [9], which computes a per-layer Block Influence (BI) score and removes individual layers with the lowest scores; `SLEB` [12], which iteratively removes the layer whose removal incurs the smallest perplexity increase on calibration data; and `Shortened LLaMA` [7], which selects layers based on one-shot perplexity evaluation. `LLM-Streamline` and `ShortGPT` are reproduced strictly following their published descriptions, while `SLEB` and `Shortened LLaMA` use the official implementations or released pruned-layer indices. In all main experiments, we prune $n = 7$ layers; results for $n \in \{9, 11\}$ are provided in Appendix **??**.

Recovery methods. After pruning, we compare the following training-free recovery methods: `None` (pruned baseline, no recovery), `Linear Patch (Diag)` and `Linear Patch (Rotate)` (LinearPatch with diagonal scaling and Hadamard rotation, respectively) [5], `Prune&Comp` [4] (channel-wise magnitude compensation applied to the surviving boundary weights), `ReplaceMe (Ls)` and `ReplaceMe (Cos)` [11] (linear transformation estimated via least squares and cosine distance,

Model	L_p/L_t	Method	ARC-E	ARC-C	HellaS	WinoG	BoolQ	OBQA	RTE	CoPa	Race	AVG
LLaMA-3-8B	0/32	Dense	77.65	53.41	79.16	72.38	81.35	45.00	69.68	89.00	40.00	67.51
	7/32	Shortened LLaMA	58.84	32.68	59.16	53.75	45.38	34.40	54.15	75.00	30.72	49.34
	7/32	LLM-Streamline (None)	39.69	28.84	33.18	55.49	38.07	29.60	57.40	60.00	24.02	40.70
	7/32	ShortGPT	56.65	42.41	64.69	71.35	65.14	32.80	67.87	75.00	34.16	56.67
	7/32	Prune and Comp	42.97	29.69	41.64	58.88	52.84	33.80	62.09	70.00	27.08	46.55
	7/32	ReplaceMe (Ls)	63.13	43.86	64.07	72.85	71.31	38.00	68.59	77.00	36.27	59.45
	7/32	ReplaceMe (Cos)	52.10	35.67	52.38	66.69	39.24	34.60	63.90	67.00	30.05	49.07
	7/32	Linear Patch (Diag)	43.86	31.74	44.14	60.85	61.25	33.60	65.34	68.00	29.09	48.65
	7/32	Linear Patch (Rotate)	50.51	34.56	49.90	63.77	57.49	33.40	67.15	68.00	30.14	50.55
	7/32	Ghost Layer (Ours)	65.82	43.69	66.55	71.27	74.34	37.20	66.43	79.00	36.56	60.10
	11/32	Shortened LLaMA	48.65	29.27	49.81	51.78	61.65	30.60	50.90	71.00	28.23	46.88
	11/32	LLM-Streamline (None)	38.17	29.69	33.04	56.67	56.15	30.20	70.04	57.00	27.18	44.24
	11/32	ShortGPT	38.17	29.69	33.04	56.67	56.15	30.20	70.04	57.00	27.18	44.24
	11/32	Prune and Comp	32.28	25.00	37.78	50.99	55.90	30.40	49.46	58.00	23.25	40.34
	11/32	ReplaceMe (Ls)	42.93	33.87	47.30	67.40	75.75	32.40	64.62	69.00	31.67	51.66
	11/32	ReplaceMe (Cos)	42.68	33.45	38.27	58.56	61.90	29.20	62.09	60.00	30.05	46.24
	11/32	Linear Patch (Diag)	46.34	35.15	47.72	61.01	71.74	31.00	70.76	68.00	30.91	51.40
	11/32	Linear Patch (Rotate)	48.15	34.56	45.04	61.40	75.47	31.60	66.79	68.00	31.29	51.37
	11/32	Ghost Layer (Ours)	48.23	34.56	53.35	69.77	75.57	32.40	65.70	71.00	32.34	53.66
LLaMA-3.1-8B	0/32	Dense	81.19	53.41	78.92	73.64	82.11	44.80	69.68	87.00	39.14	67.77
	7/32	Shortened LLaMA	61.32	33.11	59.55	54.22	43.76	35.20	51.62	77.00	31.29	49.67
	7/32	LLM-Streamline (None)	44.19	33.11	33.39	56.99	38.20	32.60	58.12	61.00	25.84	42.60
	7/32	ShortGPT	58.29	42.15	64.96	68.35	61.99	34.40	69.68	80.00	34.45	57.14
	7/32	Prune and Comp	46.25	30.89	44.22	59.12	53.91	35.00	60.29	67.00	28.61	47.25
	7/32	ReplaceMe (Ls)	64.90	43.52	63.80	71.67	69.60	37.80	71.48	77.00	37.32	59.68
	7/32	ReplaceMe (Cos)	57.03	37.29	52.14	64.25	39.36	35.80	62.82	68.00	31.10	49.75
	7/32	Linear Patch (Diag)	48.36	32.68	47.36	61.56	63.70	35.00	68.23	68.00	29.28	50.46
	7/32	Linear Patch (Rotate)	57.62	37.29	54.91	65.43	60.49	35.80	69.68	69.00	29.19	53.27
	7/32	Ghost Layer (Ours)	68.01	43.00	66.60	71.59	71.31	37.20	68.59	76.00	37.80	60.01
	11/32	Shortened LLaMA	49.54	30.38	50.05	53.91	60.80	29.20	51.26	70.00	29.38	47.17
	11/32	LLM-Streamline (None)	39.69	30.29	31.49	56.12	55.11	30.40	69.68	61.00	28.33	44.68
	11/32	ShortGPT	39.69	30.29	31.49	56.12	55.11	30.40	69.68	61.00	28.33	44.68
	11/32	Prune and Comp	38.51	27.30	40.50	53.28	62.08	29.40	53.79	57.00	25.17	43.00
	11/32	ReplaceMe (Ls)	44.53	34.04	47.23	67.40	73.55	29.80	70.76	67.00	29.95	51.58
	11/32	ReplaceMe (Cos)	43.35	32.59	37.02	56.75	58.13	29.60	68.23	62.00	31.96	46.63
	11/32	Linear Patch (Diag)	50.67	36.60	48.55	62.51	71.31	31.00	71.84	68.00	31.00	52.39
	11/32	Linear Patch (Rotate)	50.93	34.98	47.30	62.19	74.65	30.40	71.48	68.00	31.58	52.39
	11/32	Ghost Layer (Ours)	50.08	34.64	52.73	69.30	72.63	31.80	69.68	72.00	32.44	53.92
DeepSeek-R1-Distill-LLaMA-8B	0/32	Dense	65.87	42.41	74.35	67.80	82.91	41.20	69.68	89.00	41.63	63.87
	7/32	Shortened LLaMA	45.71	29.69	55.66	57.70	64.65	30.40	50.54	76.00	30.62	49.00
	7/32	LLM-Streamline (None)	49.92	35.24	46.33	61.56	51.99	32.40	57.40	70.00	27.27	48.01
	7/32	ShortGPT	49.12	37.03	55.95	63.06	77.09	34.00	74.73	71.00	33.21	55.02
	7/32	Prune and Comp	44.99	31.06	43.44	56.12	53.12	31.80	60.65	64.00	26.79	45.77
	7/32	ReplaceMe (Ls)	55.35	37.03	59.97	66.46	69.08	35.20	75.09	80.00	37.80	57.33
	7/32	ReplaceMe (Cos)	51.09	36.09	53.48	64.17	58.01	36.00	58.84	74.00	33.49	51.69
	7/32	Linear Patch (Diag)	51.09	35.32	49.59	62.27	62.78	34.60	66.43	73.00	30.05	51.68
	7/32	Linear Patch (Rotate)	54.08	36.18	53.29	64.25	72.97	33.80	62.82	72.00	32.92	53.59
	7/32	Ghost Layer (Ours)	55.81	37.03	61.30	67.25	68.47	37.00	72.92	81.00	39.43	57.80
	11/32	Shortened LLaMA	41.75	28.84	42.84	54.30	55.23	27.80	53.79	70.00	26.99	44.62
	11/32	LLM-Streamline (None)	37.08	31.66	38.72	55.17	75.20	27.40	64.26	63.00	26.22	46.52
	11/32	ShortGPT	37.08	31.66	38.72	55.17	75.20	27.40	64.26	63.00	26.22	46.52
	11/32	Prune and Comp	36.45	29.10	36.32	50.99	64.62	27.40	60.29	56.00	24.40	42.84
	11/32	ReplaceMe (Ls)	40.19	30.63	44.70	61.40	66.45	32.00	74.73	65.00	33.78	49.88
	11/32	ReplaceMe (Cos)	38.05	30.55	40.74	54.93	77.52	28.60	67.51	65.00	30.81	48.19
	11/32	Linear Patch (Diag)	45.58	34.90	43.97	58.33	77.46	31.60	68.23	64.00	29.57	50.40
	11/32	Linear Patch (Rotate)	43.69	32.76	44.22	59.19	77.34	30.40	70.04	65.00	30.72	50.37
	11/32	Ghost Layer (Ours)	42.72	32.59	48.09	64.33	75.41	31.80	74.01	66.00	34.07	52.11

Table 1: Zero-shot accuracy (%) on commonsense QA benchmarks for 7-layer and 11-layer pruning across four LLM backbones. All methods are training-free. AVG denotes the mean accuracy across all nine tasks. Here, DS-R1-LLaMA-8B is DeepSeek-R1-Distill-LLaMA-8B

respectively), and `Ghosted Layers` (ours). All baselines are reproduced using their official implementations. Our Ghosted Layers experiments are built on top of the official LinearPatch codebase [5].

Evaluation. We assess model quality along two axes: downstream task accuracy and language modeling perplexity. For accuracy, we evaluate on nine zero-shot commonsense reasoning benchmarks: `ARC-Easy` and `ARC-Challenge` [**?**], `HellaSwag` [**?**], `WinoGrande` [**?**], `BoolQ` [**?**], `OpenbookQA` [**?**], `RTE` [**?**], `COPA` [**?**], and `RACE` [**?**]. For perplexity, we evaluate on `WikiText-2` [**?**], `C4` [**?**], and `Penn Treebank` [**?**] using non-overlapping windows of length $T = 2{,}048$.

Calibration data. For layer selection and operator estimation, we use 128 randomly sampled sequences of length $T = 2{,}048$ from the C4 training split [**?**]. For SLEB, the same 128 sequences are used to evaluate perplexity at each iteration. For knowledge distillation fine-tuning, we train with AdamW at a learning rate of 1×10^{-4} for one epoch on 5,000 randomly sampled sequences of length $T = 2{,}048$ from the C4 training split [**?**].

Model	Method	7-layer WIKI	7-layer C4	7-layer PTB	7-layer PPL AVG	7-layer Acc AVG	11-Layer WIKI	11-Layer C4	11-Layer PTB	11-Layer PPL AVG	11-Layer Acc AVG
	Dense	5.47	6.71	22.51	11.56	67.51	5.47	6.71	22.51	11.56	67.51
LLaMA-3-8B	**Shortened LLaMA**	15.09	18.70	22.08	18.62	49.39	83.19	35.68	53.33	57.40	46.88
	+ Prune and Comp	12.67	17.62	20.30	16.86	49.74	231.48	141.60	313.78	228.95	41.51
	+ ReplaceMe (Ls)	68.72	58.23	121.37	82.77	43.52	17103.92	11180.54	30089.82	19458.09	37.40
	+ ReplaceMe (Cos)	14.74	18.49	21.98	18.40	49.24	55.58	43.03	63.20	53.94	45.94
	+ Linear Patch (Diag)	12.27	17.42	20.35	16.68	50.13	23.91	28.38	37.14	29.81	47.33
	+ Linear Patch (Rotate)	12.35	17.20	20.27	16.61	49.39	111.57	79.40	176.54	122.50	42.42
	+ Ghost Layer (Ours)	10.74	15.32	18.85	14.97	53.44	18.95	23.65	37.45	26.68	47.84
	LLM-Streamline	18.45	25.37	62.18	35.33	40.70	15594.89	4306.90	20481.35	13461.05	44.24
	+ Prune and Comp	14.52	20.11	55.46	30.03	46.55	449.61	375.50	745.48	523.53	40.34
	+ ReplaceMe (Ls)	30.02	26.24	52.83	36.36	59.45	133.34	78.58	380.05	197.32	51.66
	+ ReplaceMe (Cos)	121.41	153.02	203.20	159.21	49.07	2158.67	1189.44	6896.85	3414.99	46.24
	+ Linear Patch (Diag)	14.38	19.31	50.39	28.03	48.65	222.33	209.04	431.53	287.63	51.40
	+ Linear Patch (Rotate)	13.47	16.28	46.16	25.30	50.55	272.37	224.54	470.82	322.58	51.37
	+ Ghost Layer (Ours)	13.91	13.06	42.65	23.21	60.10	55.40	41.55	133.33	76.76	53.66
	ShortGPT	57.79	61.79	67.22	62.27	56.67	15594.89	4306.90	20481.35	13461.05	44.24
	+ Prune and Comp	31.20	41.84	49.95	41.00	55.18	449.61	375.50	745.48	523.53	40.34
	+ ReplaceMe (Ls)	278.00	152.34	173.75	201.36	35.89	407735776.00	265954096.00	4535126528.00	1736272133.33	38.33
	+ ReplaceMe (Cos)	79.65	66.60	151.45	99.23	56.93	130784.38	78889.44	379120.97	196264.93	39.52
	+ Linear Patch (Diag)	28.40	38.21	38.20	34.94	58.13	222.33	209.04	431.53	287.63	51.40
	+ Linear Patch (Rotate)	35.30	40.46	43.33	39.70	59.08	272.37	224.54	470.82	322.58	51.37
	+ Ghost Layer (Ours)	15.08	19.40	28.16	20.88	59.33	55.40	41.55	133.33	76.76	53.66
LLaMA-3.1-8B	Dense	6.24	8.68	10.58	8.50	67.77	6.24	8.68	10.58	8.50	67.77
	Shortened LLaMA	14.54	18.44	22.82	18.60	49.67	42.57	33.56	52.18	42.77	47.17
	+ Prune and Comp	29.41	40.46	44.26	38.04	42.60	182.71	119.57	253.54	185.27	41.22
	+ ReplaceMe (Ls)	70.03	55.41	134.37	86.60	43.84	1176.61	444.62	1827.31	1149.51	38.85
	+ ReplaceMe (Cos)	14.40	18.35	22.68	18.48	49.79	46.80	38.71	58.73	48.08	45.90
	+ Linear Patch (Diag)	12.45	17.50	20.35	16.77	41.38	224.29	136.05	314.63	224.99	40.84
	+ Linear Patch (Rotate)	12.46	17.17	20.27	16.63	43.75	232.96	180.62	343.98	252.52	52.41
	+ Ghost Layer (Ours)	10.79	15.26	18.26	14.77	–	78.81	64.56	196.16	113.18	41.33
	LLM-Streamline	2301.46	1173.53	3720.23	2398.41	42.60	32799.22	6510.32	42173.75	27161.10	44.68
	+ Prune and Comp	157.93	191.65	207.59	185.72	47.25	543.58	404.81	841.99	596.79	43.00
	+ ReplaceMe (Ls)	29.68	26.27	51.11	35.69	59.68	133.21	78.68	400.33	204.07	51.58
	+ ReplaceMe (Cos)	147.97	131.94	195.69	158.53	49.75	3469.20	1487.28	13817.19	6257.89	46.63
	+ Linear Patch (Diag)	110.93	134.68	135.50	127.04	50.46	224.32	195.13	391.71	270.39	52.39
	+ Linear Patch (Rotate)	57.33	73.48	67.38	66.06	53.27	232.96	180.62	343.98	252.52	52.39
	+ Ghost Layer (Ours)	21.35	21.52	40.56	27.81	60.01	54.98	41.31	125.73	74.01	53.92
	ShortGPT	63.42	69.64	68.58	67.21	57.14	32799.22	6510.32	42173.75	27161.10	44.68
	+ Prune and Comp	29.44	40.50	42.78	37.57	55.96	543.58	404.81	841.99	596.79	43.00
	+ ReplaceMe (Ls)	519876.94	318794.59	1379562.12	739411.22	36.47	125376232.00	59248280.00	286383744.00	157002752.00	37.48
	+ ReplaceMe (Cos)	64.41	67.57	103.02	78.33	56.43	142339.14	53814.34	325003.84	173719.11	39.55
	+ Linear Patch (Diag)	27.52	37.48	34.74	33.25	59.03	224.32	195.13	391.71	270.39	52.39
	+ Linear Patch (Rotate)	25.15	32.72	29.09	28.99	59.73	232.96	180.62	343.98	252.52	52.39
	+ Ghost Layer (Ours)	15.72	19.73	27.01	20.82	60.45	54.98	41.31	125.73	74.01	53.92
DeepSeek-R1-Distill-LLaMA-8B	Dense	13.13	19.46	22.28	18.29	63.87	13.13	19.46	22.28	18.29	63.87
	Shortened LLaMA	29.26	37.19	45.26	37.24	49.00	94.06	87.80	127.20	103.02	44.62
	+ Prune and Comp	26.82	34.22	42.28	34.44	49.29	387.30	198.67	399.84	328.60	40.08
	+ ReplaceMe (Ls)	54.69	58.63	78.63	63.98	45.94	10947.48	3420.19	10895.78	8421.15	38.16
	+ ReplaceMe (Cos)	27.60	35.12	42.89	35.20	49.75	95.91	84.17	134.11	104.73	–
	+ Linear Patch (Diag)	23.09	30.32	35.74	29.72	51.21					
	+ Linear Patch (Rotate)	23.66	30.84	36.32	30.27	51.69					
	+ Ghost Layer (Ours)	20.34	26.77	31.72	26.28	52.82	33.66	39.54	46.21	39.80	47.25
	LLM-Streamline	3083.95	1291.64	2985.68	2453.76	48.01	15094.57	4171.75	12114.49	10460.27	46.52
	+ Prune and Comp	805.43	521.87	1243.46	856.92	45.77	3061.56	1009.17	4305.52	2792.08	42.84
	+ ReplaceMe (Ls)	65.41	48.49	101.53	71.81	57.33	275.22	138.16	485.00	299.46	49.88
	+ ReplaceMe (Cos)	441.74	312.40	578.42	444.19	51.69	4059.05	2408.55	8476.96	4981.52	48.19
	+ Linear Patch (Diag)	476.26	341.89	596.51	471.55	51.68	1512.90	585.54	1998.35	1365.60	50.40
	+ Linear Patch (Rotate)	319.22	212.18	459.40	330.27	53.59	2070.12	536.66	2752.07	1786.28	50.37
	+ Ghost Layer (Ours)	45.27	38.69	59.37	47.78	57.80	115.18	75.07	181.76	124.00	52.11
	ShortGPT	343.44	157.21	906.19	468.95	55.02	15094.57	4171.75	12114.49	10460.27	46.52
	+ Prune and Comp	129.14	85.69	270.37	161.73	50.75	3061.56	1009.17	4305.52	2792.08	42.84
	+ ReplaceMe (Ls)	374225.19	1536793.62	1260787.38	1057268.73	42.78	1033096.38	1185405.75	2232597.50	1483699.88	38.89
	+ ReplaceMe (Cos)	579.23	171.66	3628.31	1459.73	54.84	24505.67	11341.19	35178.56	23675.14	44.52
	+ Linear Patch (Diag)	89.78	69.68	162.08	107.18	57.16	1512.90	585.54	1998.35	1365.60	50.40
	+ Linear Patch (Rotate)	82.07	58.87	144.75	95.23	57.33	2070.12	536.66	2752.07	1786.28	50.37
	+ Ghost Layer (Ours)	30.30	33.71	41.29	35.10	57.87	115.18	75.07	181.76	124.00	52.11

Table 2: Comparison on PPL benchmark with training-free methods over LLaMA-2-7B and LLaMA3-8B with 7 and 11 out of 32 layers pruned.

5.2 Comparison on Training-free Methods

Results on QA Benchmarks. Table D.5 reports zero-shot accuracy on nine commonsense QA benchmarks with 7 layers pruned across four LLM backbones. ReplaceMe, Linear Patch, and Ghost Layer all apply their respective recovery operators on top of LLM-Streamline as the pruning criterion. Ghost Layer consistently outperforms all competing training-free methods across all four backbones, achieving average accuracies of 58.35%, 60.10%, 60.01%, and 57.80% on LLaMA-2-7B, LLaMA-3-8B, LLaMA-3.1-8B, and DeepSeek-R1-Distill-LLaMA-8B, respectively, demonstrating that recovering the full hidden state residual at the pruning boundary yields consistent gains regardless of the underlying model architecture.

Results on PPL Benchmarks. Table **??** reports perplexity on WikiText-2, C4, and Penn Treebank (PTB) with 7 layers pruned across three LLM backbones, where each pruning criterion (Shortened LLaMA, SLEB, LLM-Streamline, and ShortGPT) is combined with all recovery methods. Ghost Layer consistently achieves the lowest perplexity across all pruning criteria and models, demonstrating that recovering the full hidden state residual at the pruning boundary yields consistent improvements in language modeling quality regardless of how the pruned layers are selected.

Model	Method	ARC-E	ARC-C	HellaS	WinoG	BoolQ	OBQA	RTE	CoPa	Race	AVG	WIKI	C4	PTB	PPL AVG
LLaMA-2-7B	Dense	74.49	46.25	75.99	68.90	77.71	44.20	62.82	87.00	39.62	64.11	5.47	6.71	22.51	11.56
	Pruned LLM	55.89	36.18	62.64	66.38	62.17	37.20	52.35	81.00	33.78	54.18	18.45	25.37	62.18	35.33
	Linear Patch (Diag) + FT	61.62	37.63	68.61	68.19	72.02	36.60	68.59	84.00	37.51	59.42	11.13	12.19	37.09	20.14
	Linear Patch (Rotate) + FT	61.95	37.97	68.59	68.27	71.83	37.00	66.79	86.00	38.28	59.63	10.91	12.07	37.28	20.09
	Ghost Layer (Ours) + FT	62.92	36.69	67.87	67.01	75.66	37.40	67.15	85.00	37.70	59.71	10.01	10.31	31.27	17.20
LLaMA3-8B	Dense	77.65	53.41	79.16	72.38	81.35	45.00	69.68	89.00	40.00	67.51	6.14	8.61	10.58	8.44
	Pruned LLM	39.69	28.84	33.18	55.49	38.07	29.60	57.40	60.00	24.02	40.70	2305.48	2106.64	4642.40	3018.17
	Linear Patch (Diag)+ FT	64.44	44.11	71.17	72.38	69.51	37.00	65.70	83.00	37.42	60.53	17.29	21.59	30.25	23.04
	Linear Patch (Rotate)+ FT	64.60	44.28	71.09	73.16	70.12	37.00	66.06	83.00	37.22	60.73	17.15	21.62	30.48	23.08
	Ghost Layer (Ours) + FT	67.59	44.11	69.19	72.22	75.54	40.00	63.90	85.00	36.65	61.58	13.00	14.96	23.03	17.00
LLaMA3.1-8B	Dense	81.19	53.41	78.92	73.64	82.11	44.80	69.68	87.00	39.14	67.77	6.24	8.68	10.58	8.50
	Pruned LLM	44.19	33.11	33.39	56.99	38.20	32.60	58.12	61.00	25.84	42.60	2301.46	1173.53	3720.23	2398.41
	Linear Patch (Diag) + FT	67.42	43.26	70.93	72.14	67.80	36.60	69.31	81.00	38.37	60.76	17.26	21.51	30.20	22.99
	Linear Patch (Rotate) + FT	67.80	43.77	70.88	72.22	67.13	36.80	69.31	81.00	37.80	60.75	17.05	21.54	30.33	22.97
	Ghost Layer (Ours) + FT	70.03	43.69	68.81	71.67	73.09	40.20	67.87	84.00	36.75	61.79	13.10	14.99	23.14	17.08
D-LLaMA-8B	Dense	65.87	42.41	74.35	67.80	82.91	41.20	69.68	89.00	41.63	63.87	13.13	19.46	22.28	18.29
	Pruned LLM	49.92	35.24	46.33	61.56	51.99	32.40	57.40	70.00	27.27	48.01	3083.95	1291.64	2985.68	2453.76
	Linear Patch (Diag) + FT	56.48	37.46	67.92	64.80	81.28	37.80	70.04	83.00	38.85	59.74	35.96	34.63	51.01	40.53
	Linear Patch (Rotate) + FT	56.48	37.54	67.72	68.11	80.15	36.40	68.95	83.00	38.37	59.64	33.64	33.47	51.39	39.50
	Ghost Layer (Ours) + FT	57.37	38.91	66.01	66.77	77.83	37.80	72.92	87.00	39.33	60.44	22.92	25.26	33.66	27.28

Table 3: Comparison on PPL and Commonsense Reasoning benchmark with SOTA post-training method LLM-Streamline and Linear Patch

		7-layer pruned				11-layer pruned			
Model	Method	GPU (%)	Speedup	Acc AVG	PPL AVG	GPU (%)	Speedup	Acc AVG	PPL AVG
LLaMA-3.1-8B	Dense	100.0	1.00×	67.77		100.0	1.00×	67.77	
	Pruned LLM	81.6	1.26×	42.60		71.1	1.48×	44.68	
	+ Prune and Comp	81.6	1.25×	47.25		71.1	1.47×	43.00	
	+ ReplaceMe (LS)	82.0	1.24×	59.68		71.5	1.45×	51.58	
	+ ReplaceMe (Cos)	82.0	1.24×	49.75		71.5	1.46×	46.63	
	+ Linear Patch (D)	82.0	1.24×	50.46		71.5	1.45×	52.39	
	+ Linear Patch (R)	82.0	1.24×	53.27		71.5	1.45×	52.39	
	+ Ghost Layer (Ours)	82.0	1.24×	60.01		71.5	1.45×	53.92	

Table 4: GPU memory usage relative to the dense model (%) and prefill speedup (×) of LLaMA-3.1-8B under varying pruning ratios. Latency is measured as the mean prefill time over 10 runs (3 warmup iterations) on a single sequence of length 2048, with `torch.cuda.synchronize()` applied before and after each forward pass.

5.3 Comparison with Post-trained Linear Patch

Table 5.2 compares Ghosted Layers against Linear Patch under both training-free and post-training settings, evaluated on nine commonsense QA benchmarks and three perplexity benchmarks. For fine-tuning, we follow the LinearPatch training protocol, adopting knowledge distillation with KL divergence against the original model's output logits as the training objective, under identical configurations across all methods (same dataset, sample count, and learning rate). Training-free Ghosted Layers already matches or outperforms Linear Patch (Diag) + FT and Linear Patch (Rotate) + FT on QA benchmarks across all four backbones, without any gradient-based optimization. With fine-tuning, Ghosted Layers + FT achieves the best performance in both QA accuracy and perplexity across all models, demonstrating that the closed-form operator provides a strong initialization that further benefits from lightweight post-training.

6 Discussion

6.1 GPU usage and Inference Latency

Latency is measured as the average prefill time over 10 runs with 3 warmup iterations on a single input sequence of length 2048, using `torch.cuda.synchronize()` to ensure accurate GPU timing.

6.2 Limitation

Ghosted Layers requires a small set of unlabeled calibration sequences to collect boundary activa-

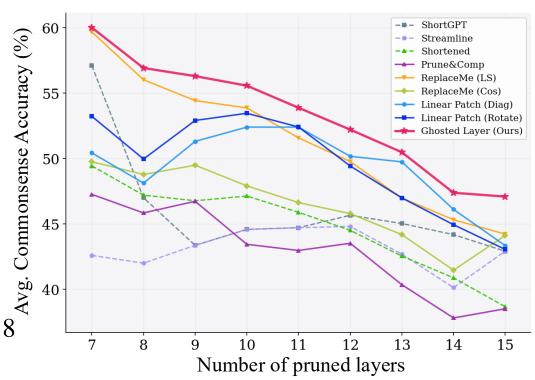

Figure 4: Average accuracy across 9 commonsense reasoning benchmarks (HellaSwag, PIQA, Wino-

tions $\mathbf{X}_{\text{pre}}$ and $\mathbf{X}_{\text{post}}$ from the unpruned model. We note that the baseline methods included in our comparison, namely LinearPatch [5], ReplaceMe [11], and Prune&Comp, similarly require forward passes through the unpruned model on calibration data to compute their respective recovery operators. To ensure a fair comparison, all methods in our experiments are evaluated under identical calibration conditions, using 128 sequences sampled from C4. Furthermore, since pruning itself requires access to the original model weights, the activations needed for recovery can be collected during the pruning process at no additional cost.

Computing the closed-form operator $\mathbf{W}^*$ requires an SVD of $\mathbf{X}_{\text{pre}} \in \mathbb{R}^{T_D \times C}$, which incurs a one-time computational cost at pruning time. We note that this cost is paid only once during the pruning phase and does not affect inference, as $\mathbf{W}^*$ is computed offline and deployed as a fixed parameter. The primary objective of Ghosted Layers is to reduce GPU memory and latency at inference time, which our efficiency measurements confirm (Table 6.1).

Pseudoinverse Error

7 Conclusion

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Appendix: Ghosted Layers

This appendix provides additional materials that complement the main paper. It includes the full theoretical analysis supporting the convergence behavior of our Variance Amplifying Regularizer (VAR), extended implementation details for all experiments, and additional qualitative and quantitative results that further illustrate the effect of VAR. The appendix is organized as follows:

- **Appendix ??**: Theoretical analysis and proofs for the proposed VAR.
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- **Appendix ??**: Theoretical analysis and proofs for the proposed VAR.
- **Appendix ??**: PyTorch implementation of the Variance Amplifying Regularizer.
- **Appendix ??**: Experimental setup, implementation details, and evaluation metrics.
- **Appendix ??**: Extended experimental results.

A Appendix

B Analyses

Theorem 2 (Ghosted Layers is the Unconstrained Optimum of Linear Patch)**.** *Let* $\mathbf{X}_{\text{pre}}, \mathbf{X}_{\text{post}} \in \mathbb{R}^{T_{\mathcal{D}} \times C}$ *and* $\mathbf{\Delta} = \mathbf{X}_{\text{post}} - \mathbf{X}_{\text{pre}}$*. Both Linear Patch and Ghosted Layers produce a repaired activation of the form* $\mathbf{X}_{\text{new}} = \mathbf{X}_{\text{pre}}\mathbf{W}$*, but differ in the structure of* $\mathbf{W}$*:*

$$\mathbf{X}_{\text{new,LP}} = \mathbf{X}_{\text{pre}} \cdot \underbrace{\mathbf{HDH}^\top}_{\mathbf{W}_{\text{LP}},\ \mathbf{W}_{\text{LP}}^\top = \mathbf{W}_{\text{LP}}} \qquad \mathbf{X}_{\text{new,GL}} = \mathbf{X}_{\text{pre}} \cdot \underbrace{(\mathbf{I} + \mathbf{X}_{\text{pre}}^\dagger \mathbf{\Delta})}_{\mathbf{W}^*,\ \mathbf{W}^* \in \mathbb{R}^{C \times C}} \tag{11}$$

$\mathbf{W}^* = \mathbf{I} + \mathbf{X}_{\text{pre}}^\dagger \mathbf{\Delta}$ *is the minimum-norm solution to the unconstrained alignment objective* $\min_{\mathbf{W}} \|\mathbf{X}_{\text{pre}}\mathbf{W} - \mathbf{X}_{\text{post}}\|_F^2$*, whereas* $\mathbf{W}_{\text{LP}}$ *searches only over symmetric matrices, making Linear Patch a constrained approximation to Ghosted Layers.*

Proof. Both methods produce $\mathbf{X}_{\text{new}} = \mathbf{X}_{\text{pre}}\mathbf{W}$.

Substituting $\mathbf{W} = \mathbf{I} + \mathbf{M}$ into the alignment objective:

$$\begin{aligned} \|\mathbf{X}_{\text{pre}}\mathbf{W} - \mathbf{X}_{\text{post}}\|_F^2 &= \|\mathbf{X}_{\text{pre}}(\mathbf{I} + \mathbf{M}) - \mathbf{X}_{\text{post}}\|_F^2 \\ &= \|\mathbf{X}_{\text{pre}}\mathbf{M} - \underbrace{(\mathbf{X}_{\text{post}} - \mathbf{X}_{\text{pre}})}_{\mathbf{\Delta}}\|_F^2 \\ &= \|\mathbf{X}_{\text{pre}}\mathbf{M} - \mathbf{\Delta}\|_F^2. \end{aligned} \tag{12}$$

Hence minimizing over $\mathbf{W}$ is equivalent to $\min_{\mathbf{M}} \|\mathbf{X}_{\text{pre}}\mathbf{M} - \mathbf{\Delta}\|_F^2$.

Since the Frobenius norm decouples across columns, it suffices to solve independently for each column $j = 1, \ldots, C$:

$$\min_{\mathbf{m}_j \in \mathbb{R}^C} \|\mathbf{X}_{\text{pre}}\mathbf{m}_j - \mathbf{\Delta}_{:,j}\|_2^2. \tag{13}$$

Taking the gradient with respect to $\mathbf{m}_j$ and setting it to zero:

$$\nabla_{\mathbf{m}_j} \|\mathbf{X}_{\text{pre}}\mathbf{m}_j - \mathbf{\Delta}_{:,j}\|_2^2 = 2\mathbf{X}_{\text{pre}}^\top(\mathbf{X}_{\text{pre}}\mathbf{m}_j - \mathbf{\Delta}_{:,j}) = \mathbf{0}, \tag{14}$$

which yields the normal equations:

$$\mathbf{X}_{\text{pre}}^\top \mathbf{X}_{\text{pre}}\, \mathbf{m}_j = \mathbf{X}_{\text{pre}}^\top \mathbf{\Delta}_{:,j}. \tag{15}$$

When $\mathbf{X}_{\text{pre}}$ has full column rank, $\mathbf{X}_{\text{pre}}^\top \mathbf{X}_{\text{pre}}$ is invertible. To handle the general rank-deficient case, let $\mathbf{X}_{\text{pre}} = \mathbf{U\Sigma V}^\top$ be the thin SVD with $\mathbf{U} \in \mathbb{R}^{T_{\mathcal{D}} \times r}$, $\mathbf{\Sigma} = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$, $\mathbf{V} \in \mathbb{R}^{C \times r}$, and $r = \operatorname{rank}(\mathbf{X}_{\text{pre}})$. The minimum-norm least-squares solution is then:

$$\mathbf{m}_j^* = \left(\mathbf{X}_{\text{pre}}^\top \mathbf{X}_{\text{pre}}\right)^{-1} \mathbf{X}_{\text{pre}}^\top \mathbf{\Delta}_{:,j} \tag{16}$$

$$= \mathbf{V\Sigma}^\dagger \mathbf{U}^\top \mathbf{\Delta}_{:,j} \tag{17}$$

$$= \mathbf{X}_{\text{pre}}^\dagger \mathbf{\Delta}_{:,j}, \tag{18}$$

where the Moore–Penrose pseudoinverse $\mathbf{X}_{\text{pre}}^{\dagger} = \mathbf{V}\boldsymbol{\Sigma}^{\dagger}\mathbf{U}^{\top}$ with:

$$\left(\boldsymbol{\Sigma}^{\dagger}\right)_{ii} = \begin{cases} 1/\sigma_i & \text{if } \sigma_i > \epsilon\sigma_1, \\ 0 & \text{otherwise.} \end{cases} \tag{19}$$

Stacking the per-column solutions over all $j = 1, \ldots, C$:

$$\mathbf{M}^* = \mathbf{X}_{\text{pre}}^{\dagger}\boldsymbol{\Delta} = \mathbf{V}\boldsymbol{\Sigma}^{\dagger}\mathbf{U}^{\top}\boldsymbol{\Delta}, \qquad \mathbf{W}^* = \mathbf{I} + \mathbf{M}^*. \tag{20}$$

Linear Patch parameterizes $\mathbf{W}_{\text{LP}} = \mathbf{H}\mathbf{D}\mathbf{H}^{\top}$, where $\mathbf{H} \in \mathbb{R}^{C\times C}$ is the Walsh–Hadamard matrix satisfying $\mathbf{H}^{\top} = \mathbf{H}^{-1}$, and $\mathbf{D} = \operatorname{diag}(d_1, \ldots, d_C)$ is a diagonal matrix.

We verify symmetry by direct computation:

$$\begin{aligned} \mathbf{W}_{\text{LP}}^{\top} &= \left(\mathbf{H}\mathbf{D}\mathbf{H}^{\top}\right)^{\top} \\ &= \left(\mathbf{H}^{\top}\right)^{\top}\mathbf{D}^{\top}\mathbf{H}^{\top} \\ &= \mathbf{H}\mathbf{D}^{\top}\mathbf{H}^{\top}. \end{aligned} \tag{21}$$

Since $\mathbf{D}$ is diagonal, $\mathbf{D}^{\top} = \mathbf{D}$, and therefore:

$$\mathbf{W}_{\text{LP}}^{\top} = \mathbf{H}\mathbf{D}\mathbf{H}^{\top} = \mathbf{W}_{\text{LP}}. \tag{22}$$

Hence $\mathbf{W}_{\text{LP}}$ is symmetric for any choice of $\mathbf{D}$, and lies in the space of $C \times C$ symmetric matrices—a subspace of dimension $\frac{C(C+1)}{2}$, strictly smaller than $C^2 = \dim(\mathbb{R}^{C\times C})$. Therefore, $\mathbf{W}_{\text{LP}}$ cannot attain $\mathbf{W}^* = \mathbf{I} + \mathbf{M}^*$ whenever $\mathbf{W}^*$ has a non-zero anti-symmetric component, i.e., whenever $\mathbf{W}^* \neq (\mathbf{W}^*)^{\top}$. $\square$

Proposition 1 (Equivalence of Minimum-Norm Solutions)**.** *Let* $\mathbf{X}_{\text{pre}}, \mathbf{X}_{\text{post}} \in \mathbb{R}^{T\times C}$ *and* $\boldsymbol{\Delta} = \mathbf{X}_{\text{post}} - \mathbf{X}_{\text{pre}}$*. Define two optimization problems:*

$$\mathbf{W}^* = \arg\min_{\mathbf{W}\in\mathbb{R}^{C\times C}} \|\mathbf{X}_{\text{pre}}\mathbf{W} - \mathbf{X}_{\text{post}}\|_F^2, \tag{23}$$

$$\mathbf{M}^* = \arg\min_{\mathbf{M}\in\mathbb{R}^{C\times C}} \|\mathbf{X}_{\text{pre}}\mathbf{M} - \boldsymbol{\Delta}\|_F^2. \tag{24}$$

Among all minimizers of each problem, taking the minimum Frobenius-norm solution yields:

$$\mathbf{W}^* = \mathbf{I} + \mathbf{M}^* = \mathbf{I} + \mathbf{X}_{\text{pre}}^{\dagger}\boldsymbol{\Delta}. \tag{25}$$

In particular, $\mathbf{W}^* = \mathbf{I} + \mathbf{X}_{\text{pre}}^{\dagger}\boldsymbol{\Delta}$ *is the minimum-norm solution to* (23)*, and coincides with* $\mathbf{X}_{\text{pre}}^{\dagger}\mathbf{X}_{\text{post}}$ *if and only if* $\mathbf{X}_{\text{pre}}^{\dagger}\mathbf{X}_{\text{pre}} = \mathbf{I}$*, i.e.,* $\mathbf{X}_{\text{pre}}$ *has full column rank.*

Proof. **Step 1: Equivalence of objectives.** Substituting $\mathbf{W} = \mathbf{I} + \mathbf{M}$ into (23):

$$\begin{aligned} \|\mathbf{X}_{\text{pre}}\mathbf{W} - \mathbf{X}_{\text{post}}\|_F^2 &= \|\mathbf{X}_{\text{pre}}(\mathbf{I} + \mathbf{M}) - \mathbf{X}_{\text{post}}\|_F^2 \\ &= \|\mathbf{X}_{\text{pre}}\mathbf{M} - (\mathbf{X}_{\text{post}} - \mathbf{X}_{\text{pre}})\|_F^2 \\ &= \|\mathbf{X}_{\text{pre}}\mathbf{M} - \boldsymbol{\Delta}\|_F^2. \end{aligned} \tag{26}$$

Hence (23) and (24) are equivalent under the bijection $\mathbf{W} = \mathbf{I} + \mathbf{M}$, with identical solution sets $\{\mathbf{W}\} = \{\mathbf{I} + \mathbf{M}\}$.

Step 2: Minimum-norm solution of (24)**.** The general least-squares solution to $\mathbf{X}_{\text{pre}}\mathbf{M} = \boldsymbol{\Delta}$ is $\mathbf{M} = \mathbf{X}_{\text{pre}}^{\dagger}\boldsymbol{\Delta} + (\mathbf{I} - \mathbf{X}_{\text{pre}}^{\dagger}\mathbf{X}_{\text{pre}})\mathbf{Z}$ for any $\mathbf{Z} \in \mathbb{R}^{C\times C}$. The minimum Frobenius-norm solution is obtained by setting $\mathbf{Z} = \mathbf{0}$:

$$\mathbf{M}^* = \mathbf{X}_{\text{pre}}^{\dagger}\boldsymbol{\Delta}. \tag{27}$$

Step 3: Minimum-norm solution of (23)**.** The bijection $\mathbf{W} = \mathbf{I} + \mathbf{M}$ is an affine isometry up to the constant shift $\mathbf{I}$. The minimizer of $\|\mathbf{W}\|_F$ subject to $\mathbf{W}$ being a minimizer of (23) satisfies:

$$\mathbf{W}^* = \arg\min_{\mathbf{W}\in\mathcal{S}} \|\mathbf{W}\|_F = \mathbf{I} + \arg\min_{\mathbf{M}\in\mathcal{S}'} \|\mathbf{I} + \mathbf{M}\|_F, \tag{28}$$

where $\mathcal{S}$ and $\mathcal{S}' = \mathcal{S} - \mathbf{I}$ are the respective solution sets. Since $\|\mathbf{I}+\mathbf{M}\|_F^2 = \|\mathbf{I}\|_F^2 + 2\langle\mathbf{I}, \mathbf{M}\rangle_F + \|\mathbf{M}\|_F^2$ and the inner product term $2\langle\mathbf{I}, \mathbf{M}\rangle_F = 2\operatorname{tr}(\mathbf{M})$ is not generally zero, the minimum-norm $\mathbf{W}^*$ over

$\mathcal{S}$ does not coincide with $\mathbf{I}+\mathbf{M}^*$ in general. However, $\mathbf{W}^* = \mathbf{I}+\mathbf{M}^*$ is the minimum-norm solution to (24) lifted back to (23) via $\mathbf{W} = \mathbf{I}+\mathbf{M}$, which is the natural parameterization consistent with the residual structure of Eq. (1).

Step 4: Relation to $\mathbf{X}_{\text{pre}}^{\dagger}\mathbf{X}_{\text{post}}$.

$$\begin{aligned} \mathbf{I} + \mathbf{X}_{\text{pre}}^{\dagger}\boldsymbol{\Delta} &= \mathbf{I} + \mathbf{X}_{\text{pre}}^{\dagger}(\mathbf{X}_{\text{post}} - \mathbf{X}_{\text{pre}}) \\ &= \mathbf{I} + \mathbf{X}_{\text{pre}}^{\dagger}\mathbf{X}_{\text{post}} - \mathbf{X}_{\text{pre}}^{\dagger}\mathbf{X}_{\text{pre}}. \end{aligned} \tag{29}$$

This equals $\mathbf{X}_{\text{pre}}^{\dagger}\mathbf{X}_{\text{post}}$ if and only if $\mathbf{I} = \mathbf{X}_{\text{pre}}^{\dagger}\mathbf{X}_{\text{pre}}$, which holds if and only if $\mathbf{X}_{\text{pre}}$ has full column rank (i.e., $\text{rank}(\mathbf{X}_{\text{pre}}) = C$). In practice, $T_{\mathcal{D}} \gg C$ for typical calibration sets, so $\mathbf{X}_{\text{pre}}$ generically has full column rank and $\mathbf{W}^* = \mathbf{X}_{\text{pre}}^{\dagger}\mathbf{X}_{\text{post}}$. $\square$

B.1 Pruning Layer Visualization

C Experimental Setup for

Figure 5: placeholder

D Additional Experiments

D.1 Data Size for Ghosted Layers

D.2 Additional Large Language Model Experiments

D.3 Experimental Settings

D.4 Not Closed Form Solution

D.5 Different Calibration Dataset

Figure 6: placeholder

Model	L_p/L_t	Method	ARC-E	ARC-C	HellaS	WinoG	BoolQ	OBQA	RTE	CoPa	Race	AVG
OlMo-2-7B	0/32	Dense	81.19	56.14	78.93	74.66	78.23	45.20	71.12	89.00	40.10	68.29
	7/32	Shortened LLaMA	71.34	43.26	68.16	65.75	65.20	41.20	62.09	79.00	33.30	58.81
	7/32	LLM-Streamline (None)	66.41	40.78	64.20	70.32	69.63	36.00	71.48	82.00	37.70	59.84
	7/32	ShortGPT	60.65	37.37	62.81	67.09	38.13	37.00	54.51	73.00	33.30	51.54
	7/32	Prune and Comp	43.56	26.37	44.68	57.30	54.68	28.80	49.10	69.00	27.08	44.51
	7/32	ReplaceMe (LS)	65.70	39.59	63.14	70.96	70.21	36.40	70.04	81.00	38.09	59.46
	7/32	ReplaceMe (COS)	66.88	40.10	63.65	70.40	67.74	36.40	73.29	81.00	38.09	59.73
	7/32	Linear Patch (D)	62.46	38.40	67.78	70.17	43.88	37.40	64.62	78.00	33.68	55.15
	7/32	Linear Patch (R)	67.34	41.64	67.01	71.03	63.94	37.20	67.87	80.00	36.27	59.14
	7/32	Ghost Layer (Ours)	67.67	41.81	63.62	70.88	70.24	37.87	70.04	80.00	37.42	59.95
	11/32	Shortened LLaMA	53.45	30.29	52.32	54.70	62.23	32.20	53.79	66.00	31.58	48.51
	11/32	LLM-Streamline (None)	50.17	33.96	52.64	67.01	56.33	33.60	78.34	71.00	30.43	52.61
	11/32	ShortGPT	36.78	25.43	33.59	51.85	38.32	26.80	50.54	66.00	25.93	39.47
	11/32	Prune and Comp	28.79	25.68	27.36	50.75	39.27	27.00	47.65	60.00	20.86	36.37
	11/32	ReplaceMe (LS)	46.34	31.91	46.22	65.90	68.04	31.40	73.65	70.00	30.33	51.53
	11/32	ReplaceMe (COS)	49.79	33.62	51.68	66.85	60.09	32.80	78.34	69.00	29.86	52.45
	11/32	Linear Patch (D)	39.86	30.55	42.43	56.99	51.19	34.80	71.84	64.00	26.99	46.52
	11/32	Linear Patch (R)	50.38	35.15	50.71	61.48	59.30	37.40	76.90	74.00	30.53	52.87
	11/32	Ghost Layer (Ours)	46.51	33.87	45.45	67.25	75.99	33.00	77.26	68.00	30.81	53.13
LLaMA-2-7B	0/32	Dense	74.49	46.25	75.99	68.90	77.71	44.20	62.82	87.00	39.62	64.11
	7/32	Shortened LLaMA	43.77	26.88	42.98	50.83	57.03	31.00	51.26	75.00	27.94	45.19
	7/32	LLM-Streamline (None)	48.61	32.76	56.15	64.48	62.17	32.80	57.40	77.00	32.25	51.51
	7/32	ShortGPT	48.61	32.76	56.15	64.48	62.17	32.80	57.40	77.00	32.25	51.51
	7/32	Prune and Comp	48.44	31.91	53.81	59.83	62.17	35.80	57.04	83.00	31.96	51.55
	7/32	ReplaceMe (LS)	50.55	35.07	54.93	64.64	65.35	33.40	58.48	74.00	35.79	52.47
	7/32	ReplaceMe (COS)	49.92	33.79	57.18	64.88	62.17	33.40	62.09	76.00	33.30	52.53
	7/32	Linear Patch (D)	55.13	34.56	57.12	63.46	62.17	35.60	55.96	78.00	35.02	53.00
	7/32	Linear Patch (R)	55.22	33.70	57.92	65.19	62.14	35.60	55.60	78.00	34.64	53.11
	7/32	Ghost Layer (Ours)	52.99	33.79	58.01	66.38	70.28	35.80	53.43	76.00	37.70	53.82
	11/32	Shortened LLaMA	44.95	25.09	42.44	51.07	47.77	30.40	54.87	72.00	27.08	43.96
	11/32	LLM-Streamline (None)	42.59	32.59	48.43	62.35	62.23	30.40	58.48	78.00	30.33	49.49
	11/32	ShortGPT	42.80	30.46	44.50	60.46	62.26	35.40	47.29	72.00	29.57	47.19
	11/32	Prune and Comp	42.68	28.16	42.79	57.70	62.26	30.00	52.71	78.00	25.93	46.69
	11/32	ReplaceMe (LS)	40.99	32.17	46.29	60.54	74.59	29.80	56.68	73.00	32.34	49.60
	11/32	ReplaceMe (COS)	38.80	32.94	46.57	57.62	62.14	30.40	60.29	74.00	31.58	48.26
	11/32	Linear Patch (D)	48.95	33.53	53.35	63.06	62.20	34.00	59.93	77.00	34.07	51.79
	11/32	Linear Patch (R)	49.28	32.51	52.73	62.98	62.17	33.20	61.73	77.00	33.88	51.72
	11/32	Ghost Layer (Ours)	49.81	31.83	49.28	65.82	73.70	33.80	57.04	72.00	33.01	51.81
LLaMA-2-13B	0/32	Dense	65.87	42.41	74.35	67.80	82.91	41.20	69.68	89.00	41.63	63.87
	7/32	Shortened LLaMA										
	7/32	LLM-Streamline (None)										
	7/32	ShortGPT										
	7/32	Prune and Comp										
	7/32	ReplaceMe (LS)										
	7/32	ReplaceMe (COS)										
	7/32	Linear Patch (D)										
	7/32	Linear Patch (R)										
	7/32	Ghost Layer (Ours)										
	11/32	Shortened LLaMA										
	11/32	LLM-Streamline (None)										
	11/32	ShortGPT										
	11/32	Prune and Comp										
	11/32	ReplaceMe (LS)										
	11/32	ReplaceMe (COS)										
	11/32	Linear Patch (D)										
	11/32	Linear Patch (R)										
	11/32	Ghost Layer (Ours)										

Table 5: Zero-shot accuracy (%) on commonsense QA benchmarks for 7-layer and 11-layer pruning across four LLM backbones. All methods are training-free. AVG denotes the mean accuracy across all nine tasks. Here, DS-R1-LLaMA-8B is DeepSeek-R1-Distill-LLaMA-8B

Model	L_p/L_t	Method	ARC-E	ARC-C	HellaS	WinoG	BoolQ	OBQA	RTE	CoPa	Race	AVG
LLaMA-3-8B	0/32	Dense										
	7/32	Shortened LLaMA										
	7/32	LLM-Streamline (None)	39.65	29.18	33.23	55.41	38.04	29.80	57.40	60.00	24.02	40.75
	7/32	ShortGPT	39.65	29.18	33.23	55.41	38.04	29.80	57.40	60.00	24.02	40.75
	7/32	Prune and Comp	42.09	29.61	41.36	59.43	51.04	33.40	59.93	68.00	27.27	45.79
	7/32	ReplaceMe (LS)	64.44	43.69	64.32	72.45	67.65	37.00	68.95	75.00	35.41	58.77
	7/32	ReplaceMe (COS)	50.93	34.73	49.71	66.14	39.02	34.00	63.18	66.00	29.67	48.15
	7/32	Linear Patch (D)	43.18	31.66	43.14	60.62	57.31	33.80	64.98	68.00	28.52	47.91
	7/32	Linear Patch (R)	51.14	34.13	49.24	63.14	57.25	34.00	67.51	67.00	29.76	50.35
	7/32	Ghost Layer (Ours)										
	11/32	Shortened LLaMA										
	11/32	LLM-Streamline (None)	38.17	29.86	32.93	56.75	56.09	30.20	70.04	57.00	27.27	44.26
	11/32	ShortGPT										
	11/32	Prune and Comp	37.25	28.16	42.86	57.14	62.81	29.20	56.32	62.00	25.07	44.53
	11/32	ReplaceMe (LS)	44.36	34.98	45.10	67.09	67.06	31.40	64.26	72.00	29.09	50.59
	11/32	ReplaceMe (COS)	42.68	33.19	37.64	57.77	61.90	29.00	62.09	57.00	29.86	45.68
	11/32	Linear Patch (D)	46.51	35.58	47.68	60.77	70.89	32.00	69.31	69.00	30.24	51.33
	11/32	Linear Patch (R)	47.81	34.64	44.03	60.85	76.09	31.60	66.43	67.00	31.96	51.16
	11/32	Ghost Layer (Ours)	49.12	34.81	51.16	68.98	77.31	31.60	66.43	74.00	32.34	53.97
LLaMA-3.1-8B	0/32	Dense										
	7/32	Shortened LLaMA										
	7/32	LLM-Streamline (None)	44.15	33.02	33.40	56.83	38.20	32.60	58.12	61.00	26.03	42.59
	7/32	ShortGPT										
	7/32	Prune and Comp	45.20	30.46	44.05	58.41	51.41	34.80	60.65	67.00	27.46	46.60
	7/32	ReplaceMe (LS)	66.67	43.09	64.23	73.01	65.72	35.00	71.12	77.00	35.41	59.03
	7/32	ReplaceMe (COS)	55.81	36.09	49.86	63.69	39.33	36.00	60.29	69.00	30.91	49.00
	7/32	Linear Patch (D)	48.06	32.76	46.46	61.40	60.34	35.20	68.23	68.00	29.00	49.94
	7/32	Linear Patch (R)	58.00	37.54	54.21	64.17	60.49	35.40	69.68	69.00	29.09	53.06
	7/32	Ghost Layer (Ours)	68.69	42.58	66.21	72.30	66.09	36.80	71.48	75.00	37.70	59.65
	11/32	Shortened LLaMA										
	11/32	LLM-Streamline (None)	39.65	30.29	31.47	56.51	55.08	30.20	69.68	61.00	28.52	44.71
	11/32	ShortGPT										
	11/32	Prune and Comp	42.47	29.27	42.88	56.20	63.79	29.00	59.93	61.00	27.37	45.77
	11/32	ReplaceMe (LS)	46.00	35.32	44.48	66.85	65.66	32.00	70.40	75.00	29.57	51.70
	11/32	ReplaceMe (COS)	43.69	32.17	36.44	57.30	58.47	29.40	68.95	62.00	31.48	46.66
	11/32	Linear Patch (D)	50.51	36.77	48.29	62.51	70.61	31.60	72.56	68.00	29.76	52.29
	11/32	Linear Patch (R)	50.80	35.15	46.60	61.88	75.29	31.00	70.04	70.00	30.72	52.39
	11/32	Ghost Layer (Ours)	50.46	34.30	50.62	68.35	74.65	32.40	67.87	72.00	33.11	53.75
DeepSeek-R1-Distill-LLaMA-8B	0/32	Dense										
	7/32	Shortened LLaMA										
	7/32	LLM-Streamline (None)	49.12	37.12	55.98	63.22	77.06	34.00	74.73	71.00	33.21	55.05
	7/32	ShortGPT										
	7/32	Prune and Comp	47.22	31.48	53.75	55.88	72.60	32.20	65.70	63.00	32.25	50.45
	7/32	ReplaceMe (LS)	54.34	37.37	60.04	66.30	73.61	33.40	72.56	81.00	40.10	57.64
	7/32	ReplaceMe (COS)	51.47	36.09	58.12	62.67	76.27	31.00	75.45	71.00	34.16	55.14
	7/32	Linear Patch (D)	54.71	36.43	60.62	64.80	81.35	33.40	74.73	72.00	36.08	57.12
	7/32	Linear Patch (R)	53.87	36.52	60.77	66.14	79.66	34.60	74.73	74.00	37.51	57.53
	7/32	Ghost Layer (Ours)										
	11/32	Shortened LLaMA										
	11/32	LLM-Streamline (None)	38.17	29.86	32.93	56.75	56.09	30.20	70.04	57.00	27.27	44.26
	11/32	ShortGPT										
	11/32	Prune and Comp	35.98	29.35	36.54	51.93	64.53	25.60	59.57	56.00	24.69	42.69
	11/32	ReplaceMe (LS)	40.49	32.34	44.05	61.56	81.80	31.80	74.37	68.00	32.15	51.84
	11/32	ReplaceMe (COS)	38.47	31.06	40.85	54.22	77.77	27.20	67.15	65.00	30.14	47.98
	11/32	Linear Patch (D)	45.12	34.22	43.12	57.85	77.40	32.00	67.87	61.00	28.61	49.69
	11/32	Linear Patch (R)	43.52	32.42	44.96	59.91	77.98	30.60	69.68	65.00	31.00	50.56
	11/32	Ghost Layer (Ours)	43.94	33.36	46.43	62.83	76.67	33.20	76.90	71.00	34.45	53.20
Olmo-2-7B	0/32	Dense										
	7/32	Shortened LLaMA										
	7/32	LLM-Streamline (None)	60.61	37.29	62.80	67.09	38.13	37.00	54.51	73.00	33.30	51.53
	7/32	ShortGPT										
	7/32	Prune and Comp										
	7/32	ReplaceMe (LS)	67.72	41.38	64.71	69.69	60.83	38.00	71.84	82.00	35.98	59.13
	7/32	ReplaceMe (COS)	66.04	41.13	65.23	68.51	44.46	37.00	70.76	77.00	35.31	56.16
	7/32	Linear Patch (D)	46.84	32.00	47.82	61.33	37.98	38.40	50.18	69.00	27.08	45.63
	7/32	Linear Patch (R)	55.93	37.88	54.90	63.30	38.75	40.20	57.04	72.00	32.15	50.24
	7/32	Ghost Layer (Ours)	65.95	42.32	62.79	69.22	70.34	39.60	71.12	81.00	37.03	59.93
	11/32	Shortened LLaMA										
	11/32	LLM-Streamline (None)	50.21	33.96	52.62	67.17	56.27	33.60	78.34	71.00	30.43	52.62
	11/32	ShortGPT										
	11/32	Prune and Comp	29.42	25.94	27.38	49.96	39.14	26.20	47.65	57.00	21.24	35.99
	11/32	ReplaceMe (LS)	47.26	32.34	46.77	66.77	60.24	32.00	70.04	71.00	30.24	50.74
	11/32	ReplaceMe (COS)	49.79	34.13	51.80	67.01	59.39	33.00	77.98	68.00	29.86	52.33
	11/32	Linear Patch (D)	40.78	32.34	44.60	56.91	48.29	35.20	70.40	66.00	28.23	46.97
	11/32	Linear Patch (R)	50.13	35.15	50.94	61.88	58.38	38.00	78.70	70.00	30.43	52.62
	11/32	Ghost Layer (Ours)	50.61	34.02	49.87	66.72	58.98	33.40	77.31	69.00	29.86	52.19

Table 6: Wiki